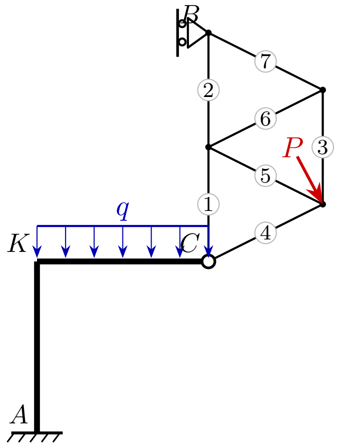
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k, reactions a, interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \;\Rightarrow\; \text{statically determinate}$$

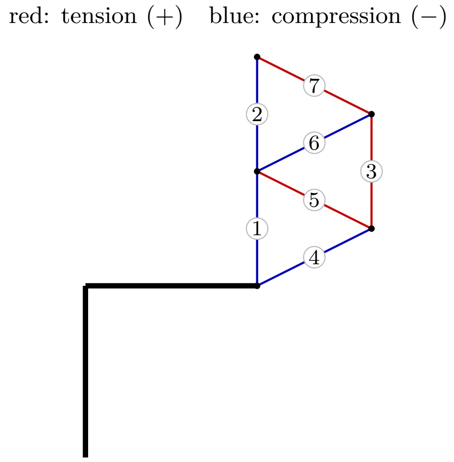
2. Support reactions and hinge forces

$A: \; R_x = 1.5, \; R_y = 24, \; M = 54 \quad B: \; R_x = -9.5, \; R_y = 0$

$C: \; H_x = -1.5, \; H_y = -15$

3. Truss member forces

Member	Force S [kN]	Type
1	-14.25	compression
2	-4.75	compression
3	9.5	tension
4	-1.68	compression
5	10.62	tension
6	-10.62	compression
7	10.62	tension



4. Section-force diagrams of the beams